LOHIT ASHWA VASWANI

AI Engineer · Data Scientist · LLM Systems Builder

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PROFESSIONAL SUMMARY

AI engineer with Distinction in MSc Data Science from University of Surrey and two years of professional experience designing, building, and deploying AI systems that move from research concept to production with measurable outcomes. Operates across many ML spectrum such as multi-agent LLM systems, RAG architectures, statistical ML, and LLM fine-tuning, with a consistent track record of translating mathematical models and data into production-grade code. Research contributions include a formal LLM benchmarking study conducted for the UK Health Security Agency currently being extended for peer-reviewed publication in PLOS Journal, a hyperparameter optimisation study achieving 94% validation accuracy on a constrained fine-tuning task, and a financial domain RAG system that drove 70% active user adoption within three months. Promoted from intern to Research Engineer at a US-based AI startup. Brings rigorous quantitative thinking, strong Python engineering fundamentals, and the discipline to own a problem from first principles through to deployment and monitoring.

TECHNICAL SKILLS

Programming: Python, SQL, NoSQL, JavaScript, C/C++
AI/ML Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, Hugging Face Transformers
Agentic AI and LLMs: LangChain, LlamaIndex, Prompt Engineering, Context Engineering, LLM Orchestration, MCP, A2A Protocol, QLoRA, PeFT
Databases & Vector Search: PostgreSQL, Oracle, MongoDB, FAISS, Pinecone
Cloud and DevOps: GCP (Vertex AI), AWS (EC2, SageMaker, Lambda), Docker, MLflow, Git, CI/CD
APIs and Backend: FastAPI, Django
Data Engineering: Apache Kafka, Databricks, Spark, ETL Pipelines, Tableau, Power BI, Selenium

PROFESSIONAL EXPERIENCE

Data Scientist Intern | CreatorOS, London, UK (Remote) **Sep 2025 – Dec 2025**

- Designed and deployed a two-stage ML recommendation system to solve CreatorOS's core product problem of unreliable brand-to-creator matching. Conducted feature engineering and statistical analysis on behavioural signals, then developed and validated a pruned RandomForest Classifier for creator segmentation and an XGBoost regression model for campaign-level ranking. Deployed via FastAPI and Docker on GCP Vertex AI at under 250ms latency. Applied MistralOCR to automate structured data extraction from unstructured client briefs directly into PostgreSQL. Production outcomes: 12% improvement in match precision (Precision@10), 10% improvement in ranking quality (NDCG@10), and 63% reduction in manual operational overhead.

Data Engineer, LLM Travel Planning PoC | Client Contract via Facebook, Karachi (Remote) **Mar 2025 – May 2025**

- Engineered Spark ETL pipelines in Databricks to consolidate multi-source travel datasets covering flights, hotels, and activities into a unified query-ready schema, resolving data quality and schema inconsistency issues across sources.
- Built a FAISS-based RAG system indexing 50,000+ travel records, improving semantic retrieval relevance and reducing incorrect itinerary suggestions during PoC evaluation.
- Developed a multi-agent LLM system with three specialised agents, a planner, a retriever, and an executor, capable of autonomously generating personalised travel itineraries that respected user-defined budget, date, and location constraints.

Research Engineer (Promoted from Intern) | Retrocausal, Redmond, USA (Remote) **Feb 2024 – Jan 2025**

Ergonomics Report Generator

- Designed and built an automated Excel reporting tool computing ergonomic risk scores including REBA, RULA, NIOSH, SNOOK, and Hand Strain from sensor data, fully integrated with the company's Django backend via POST APIs, eliminating manual spreadsheet preparation. Refactored the tool into a standalone FastAPI microservice containerised with Docker for independent deployment, horizontal scaling, and simplified CI/CD integration. Reduced report generation time by 44%, directly improving client-facing turnaround.

Model Optimisation (OpenVINO)

- Converted the production YOLOv5 pose-estimation model from PyTorch to OpenVINO IR format and applied INT8 post-training quantisation for CPU-based edge deployment without requiring GPU hardware. Achieved 2x faster inference throughput and reduced compute costs by 36% with negligible drop in keypoint detection accuracy.

Industrial Video Captioning System

- Conducted a structured comparative study fine-tuning two video-language architectures on internal assembly line footage: Timesformer+GPT-2 with temporal attention encoding and ViT+GPT-2 with frame-level encoding, analysing trade-offs between temporal modelling depth and inference efficiency to guide production model selection. The deployed captioning system reduced assembly errors by 25% and cut rework costs by 30%.

ML and Product Development Engineer Intern | Transaction Processing System, Karachi **Jul 2023 – Aug 2023**

- Developed SQL-based data extraction workflows in Oracle to prepare ATM transactional datasets of 300,000+ records, reducing data preparation time by 20% and improving query efficiency for downstream analysis.
- Built a prototype ATM withdrawal recommendation model using XGBoost with temporal and behavioural features including transaction recency, frequency, and time-of-day, reducing RMSE by 8% over baseline heuristics. Designed a lightweight end-to-end ML pipeline covering data preprocessing, feature engineering, model training, and validation, then integrated outputs into internal workflows for pilot testing across 200+ ATM users, showing 5 to 6% potential reduction in cash stockout events in simulation.
- Collaborated with product and engineering teams to align model outputs with operational requirements, contributing to a proof-of-concept for improving ATM cash management decisions.

Teaching and Academic Support | NUCES FAST University, Karachi **Aug 2022 – May 2024**

- **Student Lab Assistant, OOP (Aug 2022 to Dec 2022):** Mentored students on object-oriented principles including inheritance, polymorphism, and abstraction, guided debugging sessions, and reinforced software engineering best practices across weekly lab sessions.
- **Teaching Assistant, AI Programming (Aug 2023 to Dec 2023):** Delivered structured sessions on search algorithms, algorithmic game theory, and knowledge representation and reasoning; graded assignments and guided students in implementing AI models from scratch.

- **Teaching Assistant, Recommendation Systems (Jan 2024 to May 2024):** Guided students through collaborative filtering and neural recommender models; facilitated seminars on model evaluation, hyperparameter optimisation, and ethical considerations in recommender system design.

SELECTED PROJECTS

Automating WHO JEE Report Evaluation Using LLMs | MSc Thesis, Distinction | UK Health Security Agency | **Jun–Sep 2025**

Mistral OCR | Gemini 2.5 Flash | Qwen-3 | Magistral-Medium | BERTScore | QWK | Python

- Addressed the problem of inconsistent and labour-intensive scoring in WHO Joint External Evaluation reports by designing a fully automated LLM evaluation pipeline capable of replicating human annotation at scale. Used MistralOCR to parse and structure text from 100+ page PDFs into indicator-level JSON, then constructed a labelled dataset with rubric-based scores across Evidence Quality and Actionability dimensions.
- Benchmarked Gemini 2.5 Flash, Qwen-3, and Magistral-Medium against human annotations using Quadratic Weighted Kappa and BERTScore, demonstrating that frontier LLMs can produce consistent, reproducible evaluation of complex WHO reporting frameworks. Presented as a Proof of Concept at the UK Health Security Agency and currently extending for peer-reviewed publication in PLOS Journal.

Optimisation Methods for Mixed Discrete–Continuous LoRA Hyperparameters | **Nov 2025**

Python | PyTorch | HuggingFace Transformers | Metaheuristic Algorithms

- Developed a hyperparameter optimisation framework for LoRA-based fine-tuning of DistilBERT on a multi-class emotion classification task under strict GPU constraints. Implemented and compared five optimisation strategies: Random Search, Aquila Optimizer, Discrete Particle Swarm Optimisation with Genetic Algorithm, Blindfolded Spiderman Optimisation, and Multi-Objective Evolutionary Algorithm based on Decomposition with Differential Evolution, with consistent experimental setup across all methods. Achieved 94% validation accuracy using BSO, outperforming population-based methods while converging faster.

Fraud Detection Benchmarking: Ensemble Aggregation vs Advanced Models under Class Imbalance | **Dec 2025**

Python | Scikit-learn | Ensemble Learning | Advanced Models | Aggregation Functions

- Built a fraud detection system on 300,000+ transactions with severe class imbalance of approximately 2% fraud, applying SMOTE-Tomek for imbalance handling. Developed ensemble models combining Logistic Regression, KNN, Random Forest, and Naive Bayes with aggregation strategies including entropy-based, weighted average, threshold-aware, max probability, and trimmed mean. Benchmarked advanced models including XGBoost, Linear SVM, LDA, RIPPER, and Autoencoder+MLP using ROC-AUC, PR-AUC, precision, recall, and F1-score. Achieved 91% recall and reduced false positives from 10,000 to 800. Identified XGBoost as the best standalone model (ROC-AUC 0.97, PR-AUC 0.82) and Trimmed Mean aggregation as the best balanced ensemble.

Biomedical Abbreviation Detection and Expansion | **Apr 2025**

BioBERT | DistilBERT | PLOD-CW-25 Dataset | Token Classification | Flask | Hugging Face | Focal Loss

- Built a biomedical NLP pipeline using token-level BIO tagging to detect abbreviations and their long-form expansions in scientific literature. Benchmarked BioBERT against DistilBERT on the PLOD-CW-25 dataset, with BioBERT achieving F1 of 0.8464 due to domain-specific pretraining on PubMed and PMC corpora. Experimented with Cross-Entropy, Weighted Cross-Entropy, and Focal Loss functions, with Focal Loss improving recall on minority token classes. Augmented training data using filtered PLODv2 subsets, deployed the best model to Hugging Face, and built a Flask-based chatbot interface for real-time abbreviation expansion with custom token-level logging for model monitoring.

E-Commerce Microservices Platform | **May 2025**

FastAPI | MongoDB | Docker Compose | JWT | Sentence Transformers | React.js | Locust

- Architected a fully decoupled microservices e-commerce platform with dedicated services for user management, product catalogue, transaction processing, and personalised recommendations, each independently deployable and scalable. Implemented JWT-based stateless authentication with bcrypt password hashing and role-based access control supporting admin, buyer, and seller roles. Integrated Sentence Transformers for semantic product search enabling natural language queries beyond exact keyword matching. Orchestrated all services with Docker Compose, validated API contracts with Postman, and simulated concurrent traffic using Locust for load testing. Built a responsive React.js frontend connected via RESTful APIs.

LLM Trading Insights System (Llama-2) | BSc Final Year Project, Trade for Sight (MoU) | **Aug 2023–Mar 2024**

PyTorch | Llama-2 | LangChain | LlamaIndex | Pinecone | RAG | Chain-of-Thought Prompting

- Built under a formal MoU with Trade for Sight to solve a real financial domain problem: traders were unable to efficiently interpret complex graph-based market data, leading to slower decisions and declining platform engagement. Engineered an ETL pipeline transforming SQL-stored numerical trading data into structured natural language representations, enabling Llama-2 to perform domain-specific reasoning. Implemented Chain-of-Thought prompting to guide step-by-step decomposition of multi-variable graph data, and grounded all outputs using Pinecone vector search with LlamaIndex retrieval to prevent hallucinations. Outcomes: 30% increase in user engagement, 50% reduction in graph interpretation time, and 70% adoption among active traders within three months.

Llama-2 Augmented Mental Health Support System | **Dec 2023**

PyTorch | Llama-2 (7B) | QLoRA | PeFT | Gradio | Kaggle GPU P100 | Hugging Face

- Fine-tuned Llama-2 (7B) on the Amod mental health counselling conversations dataset to develop a virtual therapist capable of delivering empathetic, contextually appropriate responses. Applied QLoRA to quantise the model from 32-bit to 4-bit integer format, reducing GPU memory requirements from 28GB to 6GB and making fine-tuning feasible on a single Kaggle P100 GPU. Used PeFT with LoRA adapters to update less than 1% of model parameters, achieving domain adaptation at a fraction of full fine-tuning cost. Published the trained model to Hugging Face under Lohit20/Depressed_Llama-2-7b.

E-Commerce Visual Search System (CLIP + ViT) | **Oct–Nov 2023**

PyTorch | CLIP | Vision Transformer (ViT) | Pinecone | Streamlit | Colab GPU T4

- Built a multi-modal product search system allowing users to find products using either an image upload or a natural language description, eliminating reliance on precise keyword queries. Combined OpenAI CLIP with a Vision Transformer encoder to generate unified image-text embeddings, stored product embeddings in Pinecone, and retrieved the closest matches using cosine similarity. Enabled dual search modes within a single unified retrieval pipeline and deployed via a real-time Streamlit interface with results ranked by semantic similarity score.

Hybrid Movie Recommendation System (Multi-Armed Bandits + Content-Based Filtering) | **Mar 2023**

Python | Scikit-learn | Reinforcement Learning | Collaborative Filtering

- Designed a hybrid recommendation engine combining collaborative multi-armed bandits for adaptive exploration-based preference learning with content-based filtering for item-feature-driven recommendations using genre, cast, and metadata. The bandit component dynamically balanced exploration and exploitation to discover new user interests, directly addressing the cold-start problem. Evaluated recommendation diversity and accuracy metrics to validate that the hybrid approach outperformed either method independently.

Stock Price Prediction Using LSTM | **Dec 2022**

Python | Selenium | LSTM (PyTorch) | Tableau | NASDAQ

- Collected real-time stock price time-series data from NASDAQ using a Selenium-based web scraper, automating data ingestion across multiple tickers and trading sessions. Trained a multi-layer LSTM network to capture long-range temporal dependencies, achieving R^2 of 97% and MSE of 0.0680 on the held-out test set. Built an interactive Tableau dashboard presenting historical price trends, detected patterns, and 30-day forward projections for non-technical stakeholders.

Twitter Word Clustering (ETL + ML Pipeline) | **Jun 2023**

Apache Kafka | ETL | Scikit-learn | Python | Streaming Data

- Built a real-time data engineering pipeline using Apache Kafka to ingest, filter, and preprocess a continuous stream of Twitter data, handling schema variability and high-throughput ingestion reliably. Applied K-Means clustering with TF-IDF embeddings to group semantically similar tweets into coherent topic clusters, making the raw stream actionable for downstream sentiment analysis and trend monitoring.

Contextualised Course and Role Assignment (Ontology + NLP) | **Dec 2022**

Python | XML | SPARQL | RDFS | spaCy NLP

- Built a knowledge representation system that automatically assigned university courses to instructors by parsing teacher resumes with NLP entity extraction and skill classification, then matching extracted competencies against structured course requirement ontologies. Designed OWL/RDFS ontologies capturing qualification hierarchies and course prerequisites, and wrote SPARQL queries to perform logical inference and retrieve optimal course-instructor pairings via knowledge graph reasoning.

EDUCATION

- **MSc Data Science, Grade: Distinction** | University of Surrey, Guildford, UK | **Feb 2025 – Feb 2026**
  Modules: Machine Learning and Data Mining, Data Science Principles and Practices, Natural Language Processing, Cloud Computing
- **BSc Artificial Intelligence, Grade: CGPA 3.33, Three-Time Dean's List** | NUCES FAST University, Karachi, Pakistan | **Sep 2020 – Jun 2024**
  Modules: Data Structures, Algorithms, Generative AI, Computer Vision, ML, NLP, Databases, OOP
- **A-Levels** | Cedar College, Karachi, Pakistan | **Aug 2018 – Jun 2020**
  Subjects: Computer Science, Mathematics, Physics, Chemistry

CERTIFICATIONS

Orchestrating Workflows for GenAI Applications, DeepLearning.AI (Jul 2025) | Introduction to LLMs in Python, DataCamp (Apr 2024) | Generative AI Concepts, DataCamp (Mar 2024) | TensorFlow for AI, ML and Deep Learning, DeepLearning.AI (Aug 2023) | Ensemble Methods in Python, DataCamp (Nov 2022)

HONOURS AND AWARDS

- MSc Dissertation, Distinction, University of Surrey (Nov 2025)
- Three-Time Dean's List Honouree, NUCES FAST University (Spring 2021, Spring 2023, Spring 2024)
- Data Science PROCOM Competition, 7th place out of 30+ teams (Spring 2023)
- IEEE AI-Enabled TechEx Poster Competition, Finalist (Fall 2022)